GE 402 Innovative Design and Entrepreneurship II: Final Report

Neocline

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1 Executive Summary

This report presents the final deliverables and comprehensive reflections of Team Neocline's SPARK project, developed under the GE402 – Innovative Design and Entrepreneurship II course. The primary mission of this project has been to design and prototype an integrated smoking cessation system that combines hardware (SparkBox) and software (SPARK mobile app) into a behavioral support platform. The initiative aims to address one of the most persistent public health challenges: long-term smoking addiction, by leveraging user-friendly technology and data-driven strategies.

SparkBox is a compact, portable electronic ashtray embedded with a laser sensor and BLE (Bluetooth Low Energy) connectivity. It records the time and date of each cigarette disposal event and synchronizes with the SPARK mobile app to help users track patterns and receive motivational support. The SparkBox's hardware integrates efficient power management, odor-proofing, and a rechargeable battery in a pocket-sized form factor, making it suitable for real-life usage scenarios. Initially envisioned as a device with a built-in display, the SparkBox was iteratively refined to offload all visual and interaction elements to the mobile app, enhancing usability and battery longevity.

The mobile application, SPARK, was rebuilt from scratch in the second semester using React Native and Expo, following the limitations encountered with the original Flutter-based prototype. The new architecture introduced modular, screen-based navigation using 'expo-router', secure local storage for sensitive data, and a flexible design system aligned with Neocline's brand identity. Core features include user onboarding, motivational progress tracking, BLE device pairing, and a community forum designed to promote peer-based encouragement.

Significant development milestones were achieved across both hardware and software domains. BLE communication between the app and the SparkBox device was successfully established. Real-world tests confirmed a battery life exceeding 7 days, timestamp accuracy within ± 5 seconds, and reliable odor control. Meanwhile, the mobile app reached functional maturity, with a consistent UI/UX framework, navigation system, and key screens implemented. BLE data parsing and full real-time logging integration remain work-in-progress features, with detailed future plans defined in Section 7.

Beyond technical development, the team engaged in structured business planning, configuration management, and brand strategy. A comprehensive business model was developed, covering production phases, go-to-market strategy, financial forecasting, and partnership models. Brand identity was formalized under the Neocline umbrella, including visual assets, product naming, and public-facing content such as a web page. Additionally, a scalable configuration control process was introduced to support quality assurance and traceability across product versions.

Altogether, the SPARK project encapsulates the application of multidisciplinary knowledge—including embedded systems, mobile development, UX design, business modeling, and regulatory compliance—into a tangible, socially impactful solution. This report details the challenges encountered, decisions made, and lessons learned while offering a roadmap for future development and commercialization.

2 Company Organization

Sparkbox is structured as a partnership among a multidisciplinary group of dedicated students, each bringing a unique set of skills to the project. Although we are not industry experts, our diverse academic backgrounds in product development, marketing, industrial methodologies, and design enable us to approach the problem of smoking cessation with creativity and innovation. Our goal is to function as a lean, focused startup during the initial phases, with plans for scalable growth into regional and eventually international markets.

Organizational Type: We have adopted a **functional organizational structure**. Each department is responsible for a key aspect of the project, such as hardware development, software development, corporate identity, and business development. This structure allows us to streamline operations and ensure that every critical component of Sparkbox receives the attention it requires.

2.1 Product Team

The Product Team is responsible for the design, development, and refinement of Sparkbox. The team is divided into two key subgroups:

- **Hardware Product:** Led by Emre Oral, Ömer Koyuncu, and Ahmet Feyzican Güloğlu, this subgroup focuses on developing the physical components of Sparkbox, including the design, circuit integration, and enclosure.
- **Software Product:** Led by Ahmet Aybars Pektaş, the software team handles the development of the SPARK mobile application, focusing on user experience, habit tracking, and real-time data synchronization with the hardware.

2.2 Corporate Identity

Selin Uğur leads the Corporate Identity team. She is responsible for creating and maintaining the brand image of Sparkbox. This includes designing the logo, setting the brand's visual guidelines, and ensuring consistency across all marketing and communication materials.

2.3 Business Development

The Business Development team is headed by Azmi Eren Acar. His responsibilities extend beyond identifying partnership opportunities and planning growth strategies. He also oversees project management tasks, including organizing team meetings, defining Scrum goals, and ensuring that the team adheres to deadlines and deliverables.

2.4 Industrial Methodologies

Hasan Murat Talan leads the Industrial Methodologies team. His primary focus is on optimizing the development and manufacturing processes for Sparkbox to ensure high-quality standards while keeping costs manageable. Additionally, he is responsible for preparing the Quality Function Deployment (QFD) Report and the Quality Report Certification, ensuring that Sparkbox meets industry standards for reliability and user satisfaction.

2.5 Organizational Chart

Below is the organizational chart illustrating the structure of our team:

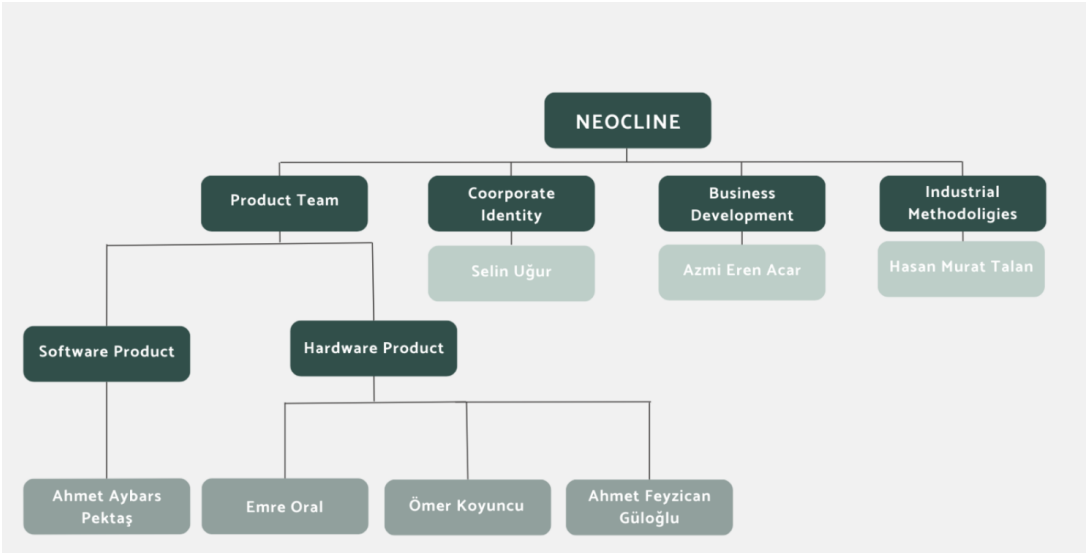

Figure 1: Organizational Chart of Neocline

3 Corporate Identity and Web Page Design

3.1 Corporate Identity

Neocline's corporate identity is built upon the principles of innovation, adaptability, and user empowerment. The brand name merges the Greek prefix "neo," meaning "new," with "cline," a term denoting gradient or change. This combination reflects the company's focus on delivering transformative, forward-thinking solutions that evolve with user needs. Typography plays a key role in conveying the brand's personality. The Neocline logotype utilizes the Prosto One font, chosen for its bold and modern style that reinforces notions of trust and reliability. For SPARK, the Anta font is employed, offering a contemporary and dynamic appearance that resonates with the product's innovative character and active mission. Color selection further enhances the brand's visual identity. Neocline's palette features #26504a, a deep green signifying balance and stability, and #b8cfc7, a lighter

green representing growth and sustainability. These shades are used in a gradient within the logotype, symbolizing seamless transformation and progress. For SPARK, the color scheme consists of #10375c, a strong and confident blue that underscores professionalism and reliability, and #f3c623, a vibrant yellow integrated as a triangular element within the letter "A" in the logotype. This detail conveys energy, upward motion, and the brand's focus on transformation and growth.

3.2 Web Page Design

The Neocline website was developed to effectively communicate the brand's mission and present its core offerings in a clear and engaging manner. It adopts a minimalist and functional design approach to ensure ease of navigation and user accessibility. The visual design aligns closely with the established corporate identity, incorporating the same fonts and color schemes for consistency. The layout features a balanced use of whitespace, structured sections, and intuitive navigation menus, allowing visitors to seamlessly explore content related to the company, the SPARK platform, and SparkBox. Content-wise, the site introduces Neocline's vision, outlines the purpose and functionality of SPARK, and provides background on the team and project development. It serves as both an informational and promotional platform, reinforcing the brand's professional image while guiding users through its innovative solutions.

4 Business Plan

This Business Plan comprises five interconnected sections: an *Organizational Plan* detailing our functional structure and team responsibilities; a *Marketing Plan* that analyzes industry trends, segments our target customers, and outlines our go-to-market and promotion strategies; an *Operations Plan* describing how SparkBox will be produced and delivered; a *Development Plan* laying out our product roadmap and risk mitigation; and a *Financial Plan* projecting revenues, expenses, and break-even dynamics over four years.

Our Business Plan revolves around commercializing SparkBox, a purpose-built, smart ashtray that automatically counts discarded cigarettes and wirelessly transmits usage data to a companion mobile app. By eliminating manual logging and enriching users with real-time insights into their smoking patterns, SparkBox occupies a unique position between pharmaceutical cessation aids and manual tracking applications. We have structured our operations to begin with an in-house prototype phase—leveraging ESP32 microcontrollers, precision laser sensors, and rapid iteration—to validate form, fit, and functionality. As demand scales, we will transition to contract manufacturing partners to achieve cost efficiencies and meet volume requirements.

On the marketing front, initial go-to-market activities will focus on direct-to-consumer online sales via our website and leading e-commerce platforms, complemented by selective retail partnerships in urban markets. Concurrently, our B2B outreach will engage corporate wellness programs and healthcare providers, offering pilot deployments that both accelerate adoption and generate clinical case studies demonstrating improved cessation

outcomes.

Our development roadmap allocates capital to finalizing the rugged, pocket-sized enclosure, optimizing Bluetooth-to-cloud connectivity, and executing user-experience refinements in the app interface. These investments, together with certification, tooling, and cloud-service expenses, have been incorporated into a four-year financial model that assumes a single product revenue stream—there is no recurring subscription fee. Under these assumptions, we project Earnings Before Tax of 0.8 million TRY in Year 1, rising to 6.5 million TRY in Year 2 as unit volumes grow, then stabilizing at approximately 4.6 million TRY in Years 3 and 4. We forecast reaching cash-flow break-even in November of Year 2, driven by a hardware selling price of 1,785 TRY and improving gross margins as production scales. Furthermore, once our formal refund policy is in place—allowing returned units to be refurbished and redeployed—the cost of goods sold is expected to decline materially, reinforcing our commitment to both profitability and sustainability.

Looking ahead, we have identified multiple exit pathways to maximize returns. In one scenario, a strategic acquisition by a major health-technology, pharmaceutical, or tobacco-cessation device company could leverage SparkBox's behavioral-data platform to augment their core offerings. Alternatively, selective licensing of our anonymized usage analytics to healthcare institutions or large insurers presents an avenue for recurring revenue. Finally, if market traction and unit economics prove exceptionally strong, we may consider an IPO on a technology-focused exchange. Each avenue depends on demonstrating sustained user growth, rigorous efficacy data, and the unique value of SparkBox's data-driven approach to smoking cessation.

5 Configuration Management and Quality Activities

5.1 Production Plan

Our production plan for SparkBox outlines three phases: Foundation & Initial Expansion, Structured Growth & Optimization, and Sustainable Scaling & Maturity, each with specific timelines, goals, key activities, and KPIs.

5.1.1 Phase 1: Foundation & Initial Expansion (0-6 Months: Summer - Fall, until Nov 2025)

Goal: To stabilize operations, launch beta, test manufacturing and supply chain, and prepare for market entry. Key Activities: Finalize beta product design with BLE and laser sensor integration, pilot manufacturing of 150 units for beta sales, begin supply chain contracts for components (approximately 1600 each), and set up the backend. Backend costs are projected at $5,000/month for the backend, $1,000/month for AWS, and $2,000/month for deployment & maintenance. Marketing is set to start in August 2025 with a budget of $15,000/month. The workforce will consist of the same team. KPIs: Pilot batch failure rate less than 5%, successful app-device sync rate greater than

or equal to 95%, and marketing engagement (CTR, CPM).

5.1.2 Phase 2: Structured Growth & Optimization (6-18 Months: Dec 2025 - Feb 2027)

Goal: To launch mass production, optimize demand, and scale infrastructure. Key Activities: Initiate mass manufacturing of over 4,000 units, full market launch with continuous customer feedback monitoring and iteration, backend load scaling for real-time sync, and implementation of an inventory and distribution management system. Cost Overview: Manufacturing cost per unit is $1600, hiring cost is $100,000, and the marketing budget will continue and may increase. Team Expansion: The developer's team will be expanded, and QA and production planners, along with a digital marketing specialist, will be hired. KPIs: Manufacturing throughput versus forecast, monthly active users (MAU), customer return rate, and average delivery time.

5.1.3 Phase 3: Sustainable Scaling & Maturity (18+ Months: Spring 2027 onward)

Goal: To stabilize production, expand the market, invest in R&D, and prepare for acquisition or scaling. Key Activities: Expand the R&D team to improve SparkBox and develop new features, increase production capacity based on demand, optimize cost per unit through volume contracts, explore new markets (e.g., Europe, partnerships), and begin discussions for strategic partnership or acquisition. Workforce: Add a strategic analyst, R&D engineers, and a business development team. KPIs: ROI tracking (20% expected by Year 2), cost per acquisition (CPA), and retention rate.

5.2 Configuration Control

The application of Configuration Control (CC) principles to the SparkBox project, emphasizing its importance for managing product evolution in a rapidly changing startup environment. Effective CC aims to ensure quality, traceability, and consistent performance across product versions.

Key aspects of the SparkBox Configuration Control Report include:

5.2.1 Configuration Identification:

Configuration Items (CIs) for SparkBox include source code (mobile app, firmware), hardware design files (CAD, PCB layouts), product specifications, Bill of Materials (BOM), user documentation, and marketing materials. Each CI will have unique identifiers and be version-controlled, utilizing Git for software and proper naming conventions for CAD files. Initial baselines will be established at prototype release, beta testing, and mass production stages.

5.2.2 Configuration Control Process:

The SparkBox team will employ a lightweight CC process. This involves using GitHub Issues or Jira to log change requests (ECRs), conducting quick impact analysis discussions for major changes, and forming a simple Change Control Board (CCB) with key team members (engineer, designer, manager). Approved changes will be documented in the issue tracker or Git pull requests, and once implemented, will be tested, verified, and included in the next release cycle.

5.2.3 Configuration Status Accounting:

The team is advised to use Git tags for identifying stable versions (e.g., v1.0-beta, v2.0-release), maintain a changelog or release notes for each version, and keep BOMs and manufacturing instructions synchronized with design file versions. Non-code documents will be managed using a shared drive or versioned folders.

5.2.4 Configuration Verification and Audit:

Verification measures include peer reviews for code and design files, automated testing for app functionality, manual inspection and checklists for physical prototypes, and comparison of 'as-designed' versus 'as-built' for hardware.

5.2.5 Simplified Workflow:

A six-step workflow is proposed: initiating changes via an issue tracker, discussing impact in team meetings, approving/rejecting changes, implementing and testing changes, updating configuration records, and communicating updates to stakeholders.

In conclusion, implementing a lightweight yet disciplined configuration control system for SparkBox is expected to reduce rework, ensure product quality, and prepare the company for scaling. Embedding CC practices early will help the startup maintain efficiency and compliance as it grows.

6 Product Development

6.1 SPARK Mobile Application

6.1.1 Overview

SPARK is a mobile application that functions as the digital companion to SparkBox. It is designed to track smoking habits, engage users through gamified and community-

based features, and provide personalized motivation to aid in smoking cessation. Initially developed using Flutter, the app was restructured and implemented in React Native with Expo for better cross-platform performance in the second semester.

6.1.2 Architecture and Tools

- **Frontend:** React Native + Expo (TypeScript)
- **Navigation:** Expo Router + React Navigation
- **Data Handling:** Expo Secure Store for encrypted local storage
- **BLE Integration:** Planned with `react-native-ble-plx`, partially implemented

6.1.3 Module Breakdown

- `welcome.tsx` – BLE pairing screen
- `login.tsx` – Authentication screen
- `info1-3.tsx` – Onboarding questions and motivation
- `analyze.tsx` – Smoking pattern visualization (incomplete)
- `community.tsx` – Posts and motivational support
- `profile.tsx` – Account and settings

6.1.4 Interface and UX

The UI follows a motivational and clean visual identity with a consistent blue-yellow theme. The mascot Sparky is used for emotional engagement. Gamification elements and screen responsiveness have been prioritized to retain user engagement.

6.1.5 Testing and Results

- BLE device pairing – **Passed**
- BLE data parsing – **Failed** (data not shown)
- Screen transitions – **Passed**
- Goal-based notifications – **Passed**
- Smoking history tracking – **Blocked by BLE parsing**

6.1.6 Development Workflow

- Version Control: GitHub with tagged versions
- Testing: Manual functional testing + BLE mock testing
- Code Review: Pull request system

6.1.7 Ethical and Legal Considerations

- All external libraries are under MIT or Expo-friendly licenses.
- No sensitive user data is transmitted externally without consent.
- GDPR and KVKK compliance are ensured via encryption and local-only storage.

6.1.8 Lessons Learned

Switching frameworks during the project presented challenges, but ultimately React Native provided better performance and design flexibility. BLE integration, although partly complete, highlighted platform limitations of Expo, guiding future development planning.

6.2 SparkBox Hardware Product

6.2.1 Overview

The Electronic Ashtray, named SparkBox, is a smart, portable device aimed at assisting users in their journey to quit smoking. It collects data on smoking habits, synchronizes with a mobile application Spark for user feedback, and promotes awareness of smoking patterns. The device integrates advanced electronic components with a user-friendly interface, providing seamless functionality to users.

6.2.2 Objective

The main objective of the SparkBox is to:

- Collect and store data regarding the user's smoking frequency (time and date).
- Provide feedback and visualizations to help users monitor and reduce their smoking habits.
- Offer odor-proof functionality to ensure a pleasant user environment.
- Enable portability and ease of use with a rechargeable power source and durable design.

6.2.3 Components

The SparkBox comprises the following components:

- ESP32 ESP-32S Wi-Fi + BLE Dual-Mode Microcontroller
- TPS63030 Buck-Boost Converter
- TP4056 Type-C and Micro USB Charging Modules
- 3.7V LiPo Battery (1050 mAh)
- IC-192B Push Button Switch
- 5mW Laser Sensor Module

6.2.4 Accomplishments and Comparison to Initial Specifications

Initially, our goal was to design a smart ashtray that could detect each cigarette disposed of, log the timestamp, and display it on a built-in screen. Over time, the design evolved to emphasize mobile integration through the SPARK app, replacing the screen with BLE data transmission to minimize complexity and power consumption.

6.2.5 Key Hardware Accomplishments

- **BLE-Based Data Transmission:** BLE functionality was implemented using the ESP32 module, enabling seamless communication with the mobile app.
- **Battery Performance:** With a 1050mAh LiPo battery and TPS63030 power management, the device operates for 7 days on a single charge.
- **Odor-Proof and Pocket-Sized:** The compact enclosure (approx. 85mm x 60mm x 25mm) includes a sealed design for odor control.
- **Charging System:** A Type-C charging module was integrated for modern convenience.

Comparison to Initial Specifications

- The originally planned on-device screen was eliminated to improve battery life and reduce complexity.
- The laser sensor was successfully integrated with the ESP32 to log cigarette disposal events.

Figure 2: Manufacturing-Ready SparkBox

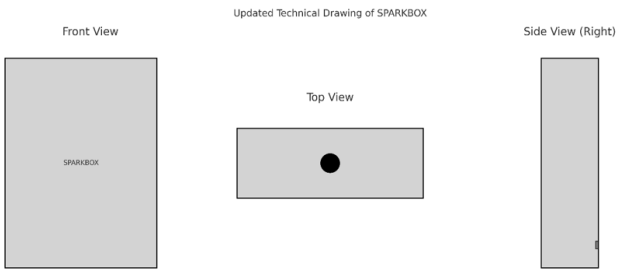

Figure 3: Technical Drawing of SparkBox

6.2.6 Test Results Summary

- **Timestamp Accuracy:** ¡ ± 5 seconds deviation.
- **BLE Sync Performance:** ¡ 3 seconds to mobile app.
- **Odor Containment:** Rated ≤ 1 on a 5-point scale.
- **Battery Life:** ≥ 7 days per charge.
- **User Notifications:** Delivered reliably within 10 seconds.

These results indicate that the prototype meets or exceeds all critical hardware goals.

6.2.7 Professional and Ethical Issues

Third-Party Hardware Components and Their Use

- **ESP32 ESP-32S**: Used as the main controller; acceptable for academic use, commercial use requires licensing.
- **TPS63030 Converter**: Ensures stable voltage; commercial release may need validation/licensing.
- **TP4056 Charger**: Widely used; requires electrical safety certification for commercial use.
- **LiPo Battery (1050 mAh)**: Rechargeable; needs certified packaging for sale.
- **IC-192B Push Button**: Freely usable; can be customized in future versions.
- **5mW Laser Module**: Used for detection; must comply with laser safety regulations if retained.

Policy Decisions to Ensure Originality and Compliance

- **Custom Enclosure and Layout:** The casing, tank, and internal arrangement were developed in-house.
- **Integration Strategy:** For efficiency, prototyping used modules. Commercial product will consolidate into a custom PCB.
- **IP Awareness:** Licensing conditions of each component were reviewed to remain compliant in academic use.
- **Scalability:** Custom circuit and board designs are planned for commercial manufacturing.

6.2.8 Conclusion

SparkBox evolved from a basic data logger into a robust, connected, and user-friendly device. Its compact size, reliable battery, and BLE communication met both user and engineering needs. The removal of the screen and emphasis on mobile app integration simplified the design while improving usability. Through prototyping and testing, the team gained experience in embedded systems, circuit integration, and real-world design constraints.

7 Future Plans

SPARK Mobile App Roadmap

Future improvements for the SPARK app include the following key directions:

- **BLE Integration Completion:** Full data parsing from SparkBox, integration with live history tracking.
- **Machine Learning Features:** Classify user smoking behavior to recommend quitting strategies.
- **Offline Support:** Local cache-based interaction without active internet or device connection.
- **Health API Integration:** Use third-party APIs to estimate fitness indicators and visualize health progress.
- **Gamification Expansion:** Introduce badges, milestones, motivational streaks, and goal-based reward logic.
- **Multi-Language Support:** Enable international rollout with language localization.
- **Security Audits:** Harden the system for commercial-grade encryption and compliance.
- **Modular Updates:** Prepare the architecture for seamless updates and feature plugin support.

8 Conclusions

The SPARK project represents the culmination of a two-semester journey combining ideation, research, development, and testing into a comprehensive smoking cessation solution. SparkBox and the SPARK mobile app were jointly conceptualized and engineered by Team Neocline to create a holistic user experience—one that blends data tracking, behavioral nudging, and peer support. The project not only fulfilled the academic criteria of the GE402 course but also laid a viable foundation for further commercial development.

The evolution from an early-stage concept with a screen-equipped ashtray to a streamlined, BLE-integrated device reflected an iterative and user-centered development process. Removing the screen simplified the power system, reduced user friction, and clarified the role of the SPARK app as the primary interface. This pivot also allowed for a lighter enclosure and optimized battery performance, which proved successful in field tests.

The mobile application development was another major area of progress. Migrating from Flutter to React Native unlocked greater design flexibility and faster development

workflows, thanks to Expo's modular infrastructure. The team implemented a complete onboarding system, motivational triggers, and BLE pairing logic within a modular and scalable navigation framework. BLE parsing remains the key remaining technical hurdle, but the architecture is fully prepared to support it once the peripheral firmware is finalized.

From a team perspective, Neocline operated under a lean, functionally organized structure, ensuring that every member took ownership of specific deliverables. The cross-functional collaboration—spanning software, hardware, branding, and business development—mirrored real-world startup dynamics and demanded accountability, agility, and frequent iteration. Through this collaborative structure, the team successfully integrated corporate identity design, QFD-based quality planning, and Agile project management.

On the professional and ethical front, SparkBox adhered to intellectual property standards by using open-source libraries and commercially unrestricted components during prototyping. Privacy and compliance concerns were proactively addressed by implementing encrypted local data storage and planning for GDPR/KVKK alignment in future releases. Future commercialization will require additional work on safety certifications, manufacturing licensing, and hardware consolidation into custom PCBs.

This project has delivered more than just a prototype—it has delivered a roadmap. The configuration control system is in place for scaling, the business plan outlines clear phases of growth, and the future feature set has been clearly prioritized. Should Neocline continue as a venture, SparkBox is well-positioned for market entry as an affordable, tech-enabled behavioral health device. Alternatively, the insights and foundations developed can be transferred to academic publications, hackathons, or health-tech partnerships.

In closing, SPARK serves not only as a capstone academic project but also as a proof of concept for how hardware-software integration and behavioral health design can converge to create scalable, impactful tools. Team Neocline exits this process with hands-on experience in product development, startup operations, and user-centered innovation—skills that will remain applicable well beyond this course.